

Technical Note

ADSS Electrical Stress Report Summary

Two types of electrical stress can cause optical and mechanical failures in all-dielectric self-supporting (ADSS) cables: tracking (i.e., dry-band arcing) and corona. When installing ADSS cables on power lines greater than 69 kV, it is important to review the structure placement of the cable and evaluate important electrical values related to tracking and corona. Electric tracking is a localized conductive path on the surface of an insulator (ADSS in this case), which can erode the cable jacket over time. A special outer jacket compound for ADSS cables may be required to combat the effects of tracking, which AFL refers to as "tracking resistant". At very high electrical stress values, the line voltage and ADSS cable placement does not allow for even a tracking resistant ADSS cable.

Corona discharge occurs when the electric field intensity exceeds the dielectric breakdown strength of air. This phenomenon can occur at the end of the metal hardware and damage the ADSS cable near that point. Corona rings are used at the end of the metal hardware to mitigate these effects.

AFL can calculate the electrical values important to determining the right fiber cable design and need of corona rings. The numerical results from the AFL electrical stress report are space potential, induced current, and the electrical field. These values must be in the acceptable ranges defined in the tables below for an ADSS cable to operate safely and effectively for its intended lifetime.

ADSS Jacket Material

Space potential and induced current are used to determine the appropriate jacket material on the ADSS cable. If either space potential or induced current values exceed standard ADSS jacket quantities, a tracking resistant jacket must be used. For space potential values above 25 kV or induced current values above 1mA, discuss with AFL alternatives for fiber cable options or ADSS structure placement.

Standard ADSS Jacket

Space Potential [kV]	< 12
Induced Current [mA]	< 0.2

Tracking Resistant ADSS Jacket

Space Potential [kV]	$12 \leq x \leq 25$
Induced Current [mA]	$0.2 \leq x \leq 1.0$

Corona Rings

Corona rings should be installed at the end of formed wire hardware when the electrical field value exceeds 14 kV/cm. This applies for both standard and tracking resistant cable jackets.

Standard and Tracking Resistant ADSS Jacket

Electrical Field [kV/cm]	> 14
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The AFL electrical stress report provides results for the electrical values of space potential, induced current, and electrical field. The report also shows graphical results of space potential and induced current values to review general cable placement and high stress areas. Note that for double-circuit systems, the circuit configuration (i.e., ABC/ABC vs. ABC/CBA) has a significant impact on electrical stress. Review this configuration in the results for accuracy.

Please contact AFL Applications Engineering for any questions.